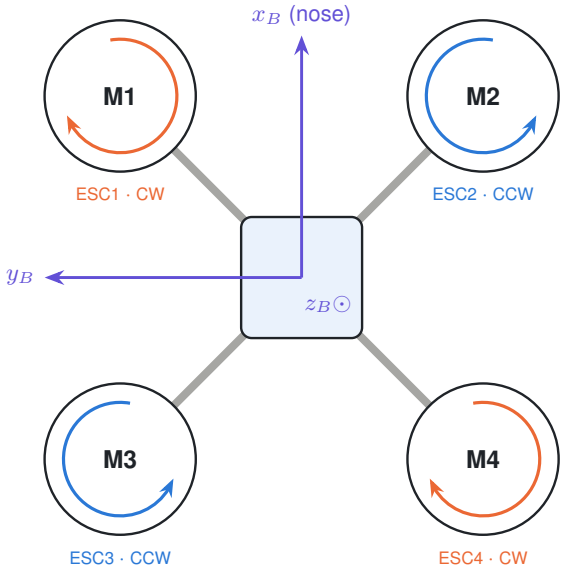
Top view, ”X” configuration (nose up)



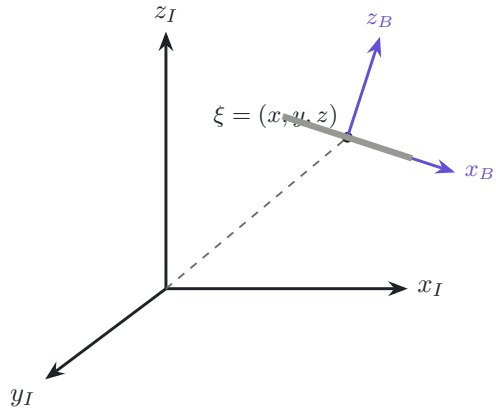
Firmware sign convention
roll $+$: left side up
pitch $+$: nose up
yaw rate $+$: nose turns right

Mixer (T throttle; r, p, y PID outputs)

$$u_1 = T - r - p + y$$
$$u_2 = T + r - p - y$$
$$u_3 = T - r + p - y$$
$$u_4 = T + r + p + y$$

diagonal pairs spin the same way,
so yaw torques cancel in hover.

Inertial frame and Euler angles (Z-Y-X)



$\eta = (\phi, \theta, \psi)$: roll about x, pitch about y, yaw about z
$R = R_z(\psi)\, R_y(\theta)\, R_x(\phi)$ maps body $\rightarrow$ inertial
model frame: z up $\Rightarrow$ firmware pitch $= -\theta$,
firmware yaw rate $= -r$